Practical No : 12

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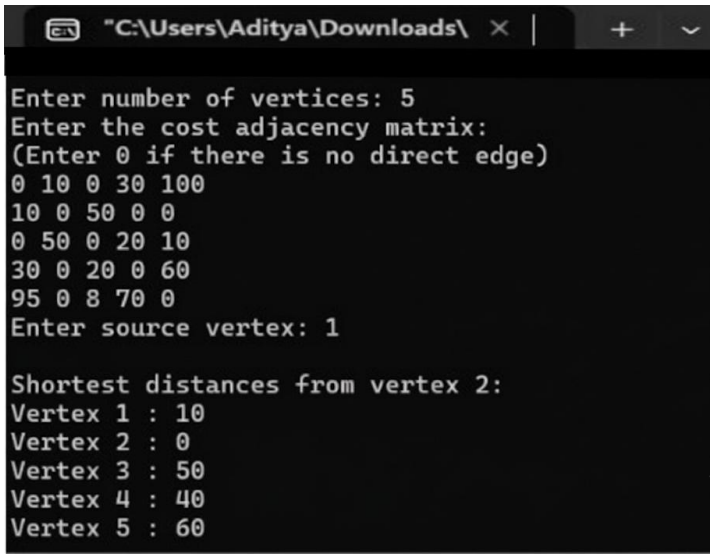
Q.12) Write a C program for Dijkstra's Algorithm.

```
#include <stdio.h>
#include <limits.h>
#define MAX 20
int main()
{
   int n, graph[MAX][MAX];
   int distance[MAX], visited[MAX];
   int i, j, source;
   int min, u;
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   printf("Enter the cost adjacency matrix:\n");
   printf("(Enter 0 if there is no direct edge)\n");
   for (i = 0; i < n; i++) {
      for (j = 0; j < n; j++) {
         scanf("%d", &graph[i][j]);
      }
   }
   printf("Enter source vertex: ");
   scanf("%d", &source);
   source;
   for (i = 0; i < n; i++) {
      distance[i] = INT_MAX;
      visited[i] = 0;
   }
```

```
    distance[source] = 0;
    for (i = 0; i < n - 1; i++) {
        min = INT_MAX;
        u = -1;
        for (j = 0; j < n; j++) {
            if (!visited[j] && distance[j] < min) {
                min = distance[j];
                u = j;
            }
        }
        if (u == -1)
            break;
        visited[u] = 1;
        for (j = 0; j < n; j++) {
            if (!visited[j] &&
                graph[u][j] != 0 &&
                distance[u] != INT_MAX &&
                distance[u] + graph[u][j] < distance[j]) {
                distance[j] = distance[u] + graph[u][j];
            }
        }
    }
    printf("\nShortest distances from vertex %d:\n", source + 1);

    for (i = 0; i < n; i++) {
        if (distance[i] == INT_MAX)
            printf("Vertex %d : INF\n", i + 1);
        else
            printf("Vertex %d : %d\n", i + 1, distance[i]);
    }
    return 0;
}
```

Output:


"C:\Users\Aditya\Downloads\
Enter number of vertices: 5
Enter the cost adjacency matrix:
(Enter 0 if there is no direct edge)
0 10 0 30 100
10 0 50 0 0
0 50 0 20 10
30 0 20 0 60
95 0 8 70 0
Enter source vertex: 1
Shortest distances from vertex 2:
Vertex 1 : 10
Vertex 2 : 0
Vertex 3 : 50
Vertex 4 : 40
Vertex 5 : 60